

100 DSA Interview Questions

Beginner → Advanced | JavaScript / MERN Developer Interview Preparation

Recommended Learning Order

Arrays → Strings → Searching → Sorting → Linked List → Stack → Queue → Recursion → Trees → BST → Graph → Dynamic Programming

1. Arrays — Questions 1 to 20

1. Find the largest element in an array.
2. Find the second largest element.
3. Find the smallest element in an array.
4. Reverse an array.
5. Check if an array is sorted.
6. Remove duplicates from a sorted array.
7. Move all zeros to the end.
8. Find the missing number from 1 to N.
9. Find duplicate elements in an array.
10. Find the frequency of each element.
11. Find the sum of all array elements.
12. Find the intersection of two arrays.
13. Find the union of two arrays.
14. Find the maximum difference between two elements.
15. Find the maximum subarray sum — Kadane's Algorithm.
16. Find all pairs whose sum equals a target.
17. Solve Two Sum.
18. Rotate an array by K positions.
19. Find the majority element.
20. Find the subarray with a given sum.

2. Strings — Questions 21 to 35

21. Reverse a string.
22. Check whether a string is a palindrome.
23. Count vowels and consonants.
24. Find duplicate characters.
25. Find the first non-repeating character.
26. Count frequency of each character.
27. Check whether two strings are anagrams.

- 28. Remove duplicate characters from a string.
- 29. Find the longest word in a sentence.
- 30. Reverse words in a string.
- 31. Find the longest common prefix.
- 32. Check whether one string is a rotation of another.
- 33. Find the longest substring without repeating characters.
- 34. Compress a string.
- 35. Check whether a string contains only unique characters.

3. Searching & Sorting — Questions 36 to 48

- 36. Implement Linear Search.
- 37. Implement Binary Search.
- 38. Find the first occurrence of an element.
- 39. Find the last occurrence of an element.
- 40. Find the insertion position of an element.
- 41. Implement Bubble Sort.
- 42. Implement Selection Sort.
- 43. Implement Insertion Sort.
- 44. Implement Merge Sort.
- 45. Implement Quick Sort.
- 46. Find the Kth smallest element.
- 47. Find the Kth largest element.
- 48. Find an element in a rotated sorted array.

4. Linked List — Questions 49 to 60

- 49. Create a singly linked list.
- 50. Traverse a linked list.
- 51. Insert a node at beginning.
- 52. Insert a node at end.
- 53. Insert a node at a given position.
- 54. Delete a node from a linked list.
- 55. Reverse a linked list.
- 56. Find the middle of a linked list.
- 57. Detect a cycle in a linked list.
- 58. Find the starting point of a cycle.
- 59. Merge two sorted linked lists.
- 60. Remove the Nth node from the end.

5. Stack & Queue — Questions 61 to 70

- 61. Implement Stack using an array.
- 62. Implement Queue using an array.
- 63. Implement Stack using Linked List.
- 64. Implement Queue using Linked List.
- 65. Implement a Queue using two Stacks.
- 66. Implement a Stack using two Queues.
- 67. Check balanced parentheses.
- 68. Find the next greater element.
- 69. Implement Min Stack.
- 70. Reverse a Stack.

6. Recursion & Backtracking — Questions 71 to 78

- 71. Calculate factorial using recursion.
- 72. Calculate Fibonacci using recursion.
- 73. Find the sum of digits recursively.
- 74. Reverse a string using recursion.
- 75. Generate all subsets of an array.
- 76. Generate all permutations of a string.
- 77. Solve the N-Queens problem.
- 78. Solve a Sudoku.

7. Trees — Questions 79 to 88

- 79. Create a Binary Tree.
- 80. Inorder traversal.
- 81. Preorder traversal.
- 82. Postorder traversal.
- 83. Level-order traversal.
- 84. Find the height of a binary tree.
- 85. Count nodes in a binary tree.
- 86. Find the maximum value in a binary tree.
- 87. Check whether two trees are identical.
- 88. Check whether a binary tree is balanced.

8. Binary Search Tree — Questions 89 to 93

- 89. Search an element in a BST.
- 90. Insert an element into a BST.
- 91. Delete an element from a BST.

- 92. Find the minimum and maximum in a BST.
- 93. Find the Lowest Common Ancestor in a BST.

9. Graphs — Questions 94 to 97

- 94. Implement a graph using an adjacency list.
- 95. Implement BFS.
- 96. Implement DFS.
- 97. Detect a cycle in a graph.

10. Dynamic Programming & Advanced — Questions 98 to 100

- 98. Solve Climbing Stairs using DP.
- 99. Solve 0/1 Knapsack.
- 100. Find the Longest Common Subsequence (LCS).

Top 20 Most Important Questions

If you have limited time, prioritize these problems first.

1. Two Sum
2. Reverse Array
3. Maximum Subarray — Kadane
4. Move Zeros
5. Remove Duplicates
6. Missing Number
7. Majority Element
8. Binary Search
9. First/Last Occurrence
10. Rotated Sorted Array
11. Reverse Linked List
12. Middle of Linked List
13. Detect Linked List Cycle
14. Merge Two Sorted Lists
15. Valid Parentheses
16. Next Greater Element
17. Binary Tree Traversals
18. Maximum Depth of Binary Tree
19. BFS/DFS
20. Climbing Stairs / 0-1 Knapsack

Recommended solving method: For every problem, first understand the brute-force approach, then optimize it. Write the JavaScript solution, perform a dry run, and always identify time and space complexity.